

Remote Access and Physical Security

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Implementing Remote Access at Big Tire Locations

To support employees and trusted business partners across all Big Tire locations to work effectively, Big Tire will implement a secure and flexible remote access solution. This solution will allow authorized users and devices to connect to key applications such as logistics and accounting. The goal is to improve both security and performance, especially for users working remotely or on mobile devices.

1. Virtual Desktop Infrastructure (VDI):

- Through Virtual Desktop Infrastructure, employees will gain secure access to data center-hosted applications, resolve VPN performance problems, and achieve consistent user experiences.

2. Secure HTTPS Access:

- Remote users will have secure access to critical applications like logistics and accounting software through an encrypted HTTPS connection, which maintains data security.

3. Multi-factor Authentication (MFA):

- Multi-factor Authentication (MFA) must be used for all remote access to enhance security. It validates users with primary credentials along with an additional verification factor, such as an OTP or authentication app.

4. Remote Access Gateway:

- The remote access gateway authenticates users to ensure only authorized personnel can connect to applications.

5. Performance Optimization:

Performance optimization includes load balancing and application acceleration to deliver seamless access to bandwidth-intensive applications.

Implementing Physical Security at Big Tire Locations

Big Tire will improve physical security across all Big Tire locations by deploying smart cards and programmable locks for offices and garages, ensuring consistent access control and enhanced protection of company assets. This will help ensure that only authorized individuals can enter sensitive areas.

1. Smart Card Access Control:

- Employees will use smart cards to control their entry to office buildings and garage spaces. Specific permissions will be programmed into each card to restrict access to authorized personnel only for entering restricted areas. Central management of the system enables straightforward updates.

2. Programmable Locks:

- The company will place programmable locks on critical doors and protected areas. The locks will enable detailed access management for specific areas while allowing permission modifications according to requirements.

3. Surveillance Systems:

- CCTV cameras will be installed at key points throughout the property to oversee entrances, loading docks, and additional high-risk zones. These cameras will improve security measures by offering uninterrupted surveillance combined with instantaneous notifications about suspicious activities.

4. Employee Training:

- The company will educate employees on operating access control systems and identifying suspicious activities. The training program will establish security awareness throughout the entire company.

5. Remote Monitoring and Alarms:

- Alarm systems in each facility will operate in conjunction with remote monitoring services. Security personnel will receive immediate alerts when a security breach happens.

6. Access Audits:

- Scheduled audits will examine access logs to verify the effectiveness and currency of security measures. Conclusion: The physical security measures Big Tire implements will protect its facilities while decreasing risks related to unauthorized entry and creating a safer workplace environment.

High Level Plan for Cost-Effective Multi Factor Authentication for Websites

To protect web-based systems and sensitive data, Big Tire will implement multi factor authentication using authenticator apps, biometrics, and tokens, depending on the user's role and device. The focus will be on keeping the process secure, easy to use, and affordable for Big Tire.

1. Authentication methods

- Authenticator Apps (Free) - Microsoft or Google security
- Hardware Security Keys (High Risk Users) - YubiKey- \$60 per unit
- Biometrics (Built in)

2. Implementation and Costs

- Total Annual Budget starting \$100,000-\$150,000
 - Kansas (500 Employees) - \$50,000
 - Minneapolis (200 Employees) - \$20,000
 - Memphis (150 Employees) - \$15,000
 - Reno (175 Employees) - \$17,000
 - El Paso (250 Employees) - \$22,000
- Hardware security keys - for around 500 high risk users at \$60, \$30,000 will go towards these security keys for all locations
- IT support and training will cost around \$40,000 for all locations.
- New Software at all locations with the remaining budget, around \$50,000

- Every year forward we will evaluate the budget needs, if more security is needed, we can increase, and if less is needed, we will cut down.

3. Justification

- Prevent cyber-attacks to the maximum ability with a reasonable budget
- Balance cost, user convenience and company security

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Student Contributions

Student	Contributions
Ahmad Barakat	- high-level plan for cost-effective multi-factor authentication for websites - Meeting final submission requirements
Alejandra Garcia	- Descriptions for implementing remote access, physical security, and multi-factor authentication plan
Makayla Atherton	- High-level plan for implementing physical security at Big Tire locations
Andrew Pierce	- High-level plan for implementing remote access to applications hosted in the data center
Braxton McCarty	- No contribution